



A conversational agent based on contingent teaching model to support collaborative learning activities: impacts on students' learning performance, self-efficacy and perceptions

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Abstract

To promote deeper cognitive interactions and positive socio-emotional interactions among group members, thereby achieving high-quality collaborative outcomes, researchers have endeavored to develop conversational agents (CAs) that provide adaptive support to small groups. However, existing CAs for supporting collaborative learning struggled to integrate and analyze multi-source learning data throughout the collaborative learning process and to offer comprehensive and personalized scaffolding based on diagnostic results. To address these issues, this study designed a CA named CollaBot, based on the contingent teaching model, which has been used in the past to guide teachers on how to provide adaptive scaffolding to small groups. CollaBot integrates AI technologies, including retrieval-based models, generative AI models, and retrieval-augmented generation techniques, to offer adaptive cognitive, metacognitive, and social scaffolding to groups engaged in online collaborative learning. A randomized controlled experimental design was employed, recruiting 78 undergraduate students who were randomly divided into two groups: the experimental group (n=39) utilized a co-writing platform with CollaBot, while the control group (n=39) used the platform with task scripts. Results indicate that students supported by CollaBot demonstrated significantly better learning performance. In addition, both CollaBot and task scripts significantly enhanced students' self-efficacy for writing. Furthermore, analysis of the interview data revealed both positive perceptions about CollaBot, such as aiding group members in regulating their own and the group's learning processes and supporting the development of writing skills, as well as negative perceptions, including causing anxiety and providing ambiguous feedback. This research provides guidance for the design of CAs and offers insights into harnessing hybrid intelligence between teachers and GAI to support collaborative learning.

Keywords Collaborative learning · Generative AI · Conversational agents · Retrieval-augmented generation · Contingent teaching model

Introduction

In online learning, students have been encouraged to work together, exchange information, and collaborate on their learning tasks (Kozlov & Große, 2016). During online collaborative activities, students often encounter varied cognitive and socio-emotional challenges (Li et al., 2023), such as insufficient prior knowledge (Baytelman et al., 2020) and negative socio-emotional interactions (Mänty et al., 2020), which may impede the production of high-quality learning outcomes. Previous studies have suggested that students require the cognitive, metacognitive, social scaffolding to achieve productive collaboration (Ouyang et al., 2021; Zheng et al., 2019). Cognitive scaffolding typically involves providing prompts to guide students in knowledge construction or offering direct guidance to address students' confusion (Rojas-Drummond et al., 2013; Van de Pol et al., 2014). Metacognitive scaffolding involves guiding students in planning, monitoring, and reflecting on the collaborative learning process (Ouyang et al., 2022; Zheng et al., 2022). Social scaffolding typically involves fostering a positive emotional atmosphere within groups and facilitating social interactions among students (Ouyang et al., 2022; Zhu & Lin, 2023). In the past, scaffolding was typically provided by teachers through dialogue (eg. CaLor et al., 2022; Rojas-Drummond et al., 2013; Van de Pol et al., 2014; Zhu & Lin, 2023). According to the contingent teaching model proposed by Van de Pol et al. (2012), the process of teacher scaffolding small groups often involves two core steps: diagnosing the group learning situation, and then providing adaptive support accordingly. Although this approach has been proven to be effective, teachers often encounter obstacles in monitoring all groups and offering timely support (Van de Pol et al., 2012; Van Leeuwen et al., 2015). Addressing this issue, researchers highlighted that conversational agents (CAs), as a kind of scaffolds, have the potential to automatically provide adaptive and personalized support to collaborative groups (Hayashi, 2019; Okita & Clarke, 2021), thereby effectively alleviating the teaching burden on teachers (Puntambekar, 2022).

Similar to the process by which teachers provide scaffolding to collaborative groups, CAs also need to first diagnose the group's collaborative learning situation. The accuracy of the diagnostic results is crucial, as it directly influences the effectiveness of the scaffolding provided (CaLor et al., 2022; Van de Pol et al., 2010). Researchers have emphasized the importance of integrating data from multiple dimensions for effective diagnosis, including collaborative conversations, group learning behaviors, and group outcomes (Van de Pol et al., 2012; Van de Pol et al., 2014), which serves as the foundation for offering adaptive scaffolding to learners. However, existing CAs designed to support collaborative learning primarily rely on unidimensional data sources, typically focusing on dialogues for their diagnostic processes (e.g. Darmawansah et al., 2024; Dyke et al., 2013; Guo et al., 2024; Martha et al., 2023; Nguyen, 2023; Tegos et al., 2016). This limitation means that current CAs are unable to comprehensively understand the group's collaborative learning situation, making it difficult for them to provide sufficiently adaptive support. Meanwhile, numerous studies have employed multimodal learning analytics methods, utilizing manual coding or machine learning techniques to analyze multiple data sources, such as student dialogues and clickstreams, within collaborative learning processes (eg. Du et al., 2022; Ouyang et al., 2023; Xing et al., 2022). These studies have laid the foundation for designing CAs with enhanced diagnostic capabilities, enabling them to analyze multimodal data in real-time to comprehensively understand the dynamics of learner groups.

After diagnosing the group's learning situation, the CAs will provide different types of scaffoldings to the small group. Most CAs in the studies can provide cognitive

scaffolding, helping learners gather and organize information, and assisting them in building knowledge based on others' opinions (e.g. Adamson et al., 2014; Darmawansah et al., 2024; Dyke et al., 2013; Guo & Li, 2024; Tegos et al., 2014, 2016; Wang et al., 2011). Some CAs in the research can offer metacognitive scaffolding, facilitating students' reflection on the learning process and encouraging them to make adjustments through questioning or suggestions (e.g. Hayashi, 2020; Martha et al., 2023; Yilmaz & Yilmaz, 2019). Fewer studies designed CAs that provide social scaffolding to groups, supporting positive socio-emotional interactions among collaborative groups (eg. Nguyen, 2023). Although previous researchers highlighted the significance of providing metacognitive, cognitive, and social scaffolding to collaborative groups (Ouyang et al., 2022; Rojas-Drummond et al., 2013; Van de Pol et al., 2014), there was currently limited research on designing CAs with the capability to provide these three types of scaffolding.

In addition, most CAs for collaborative learning utilized retrieval-based techniques (e.g. Nguyen, 2023; Tegos et al., 2016; Wang et al., 2011; Xie et al., 2021), which meant that the scaffolding provided by CAs is predefined, making it challenging to meet the personalized needs of learners. Recent advancements in generative AI, exemplified by ChatGPT (Chen et al., 2024), present an opportunity to overcome some limitations of CAs for collaborative learning. Generative AI demonstrated groundbreaking language understanding and generation capabilities (Adeshola & Adepoju, 2023), implying that generative AI-based CAs can effectively comprehend collaborative conversations and generate adaptive scaffolding. A small number of studies have already employed generative AI-based CAs to support collaborative learning, validating their effectiveness in scenarios such as peer feedback (Guo et al., 2024) and collaborative argumentation (Darmawansah et al., 2024). However, CAs in these studies still face challenges in analyzing multi-source data to provide cognitive, metacognitive, and social scaffolding to groups.

At the same time, while researchers generally acknowledge the powerful capabilities of generative AI (Chen et al., 2024), they have also raised concerns that students may become overly dependent on generative AI (Chan & Hu, 2023), potentially hindering the development of independent thinking skills (KikaLishvili, 2023), and may generate incorrect content that could mislead students (Barrot, 2023). To address these issues, retrieval-augmented generation (RAG) techniques offer potential solutions. The core principle of RAG is to retrieve knowledge relevant to the user's query and provide it to the large language model to effectively control the generated content (Lewis et al., 2020). Recently, educational researchers have advocated for the application of RAG techniques in designing CAs to more effectively facilitate learning (Guo & Li, 2024; Lo et al., 2024). However, there is still a lack of studies that have designed CAs based on RAG and conducted empirical research to validate their effects (Guo & Li, 2024).

Pondering existing research gaps, this study integrated various technologies, including generative AI models, retrieval-based models and RAG method, to design and develop a CA named CollaBot. CollaBot had the capability to adaptively provide cognitive, metacognitive, and social scaffolding to collaborative groups by integrating diverse data, including group conversations, group writing content, and group writing behaviors. Furthermore, a randomized controlled experimental design was employed to examine the effects of CollaBot on students' learning performance and self-efficacy for writing, as well as to investigate their perceptions of CollaBot.

Literature review

Teacher scaffolding on collaborative learning

Common teacher scaffolding that supports collaborative learning includes three types, which are cognitive, metacognitive, and social scaffolding (Ouyang et al., 2022). In terms of cognitive scaffolding, teachers usually support the group's knowledge construction and problem-solving by providing hints or direct instruction (Ouyang et al., 2022; Rojas-Drummond et al., 2013; Zheng et al., 2022; Zhu & Lin, 2023). Researchers classify such cognitive scaffolding into different types based on the degree of control (Ouyang et al., 2022; Van de Pol et al., 2019). Low-control cognitive scaffolding typically involves open-ended questions or hints indicating the direction for learners to solve problems (Van de Pol et al., 2019). Meanwhile, high-control cognitive scaffolding provides learners with explicit guidance on what to do or directly offers answers to questions (Ouyang et al., 2022). Regarding metacognitive scaffolding, teachers assist students in planning, monitoring, and reflecting (Ouyang et al., 2022; Rojas-Drummond et al., 2013; Zheng et al., 2022). Specifically, teachers guide groups in setting goals and developing plans based on those goals at the outset of activities (Zheng et al., 2022). They provide reminders when groups deviate from the planned progress (Rojas-Drummond et al., 2013) and prompt group members to reflect on collaborative tasks (Zheng et al., 2022). Considering social scaffolding, teachers typically strive to maintain a positive socio-emotional atmosphere within the small groups (Ouyang et al., 2022) and establish community norms (Zhu & Lin, 2023). Specifically, teachers provide encouragement when group members are discouraged (Rojas-Drummond et al., 2013; Zheng et al., 2022) and remind them to engage actively in social interactions (Zhu & Lin, 2023). All these three types of teacher scaffolding are proven to facilitate collaborative learning, giving essential insights on what to scaffold for teachers in collaborative learning context.

In addition to discussing what to scaffold in collaborative learning, researchers also focus on how to scaffold collaborative learning. Van de Pol et al. (2012) proposed the Contingent Teaching Model (CTM), outlining the concrete steps for teachers to scaffold small groups. According to the CTM, the teacher starts by diagnosing a student's current understanding through methods like asking diagnostic questions or reviewing their work. The teacher verifies their own understanding to establish a shared comprehension. Subsequently, the teacher offers support tailored to the diagnostic information collected. Lastly, the teacher confirms the student's updated understanding. The subsequent study conducted by Van de Pol et al. (2014) employed CTM to guide teachers in scaffolding small groups, finding CTM can optimize the quality of teacher scaffolding. In addition, Behzadi Soufiani et al. (2022) employed CTM to improving Iranian EFL learners' essay writing, finding CTM could promote learners' essay writing performance. Furthermore, Calor et al. (2022) have made enhancements to the CTM, providing guidance to instructors on how to offer support to the whole group. In summary, research on teacher scaffolding has established a theoretical foundation to elucidate what to scaffold and how to scaffold in collaborative learning, and provided abundant empirical evidence to illustrate the positive effects of teacher scaffolding (Ouyang et al., 2022; Rojas-Drummond et al., 2013; Van de Pol et al., 2014; Van de Pol et al., 2019; Zheng et al., 2022; Zhu & Lin, 2023). However, researchers also point out that teachers face significant challenges when scaffolding small groups, as they often struggle to monitor all groups and offer timely support (Van de Pol et al., 2012; Van Leeuwen et al., 2015). Okita and Clarke (2021) advocated that future studies should

focus on designing and developing automated scaffolds like conversational agents, aiming to provide adaptive and personalized support to small groups.

Conversational agents for collaborative learning

Conversational agents, also known as chatbots, are capable of automatically providing adaptive support to learners through dialogues with them in collaborative learning (Hayashi, 2019; Okita & Clarke, 2021). Researchers have designed various CAs to support collaborative learning and conducted a variety of empirical studies to explore their effects (e.g. Adamson et al., 2014; Darmawansah et al., 2024; Dyke et al., 2013; Guo et al., 2024; Hayashi, 2019; Martha et al., 2023; Nguyen, 2022, 2023; Tegos et al., 2014, 2016; Wang et al., 2011). Designing CAs involves formulating automated diagnostic and scaffolding strategies, both deemed crucial parts of the scaffolding process (Van de Pol et al., 2010, 2012). Regarding diagnostic strategies, most CAs in relevant studies diagnose only based on conversation data. For example, CAs in the study of Nguyen (2023) and Tegos et al. (2016) analyze collaborative dialogues, assess the activation status of learners' knowledge points, and subsequently provide adaptive feedback based on the diagnostic results. Guo et al. (2024) designed a generative AI-based CA to support peer assessment. This CA is capable of analyzing the feedback provided by each student, helping them to offer higher-quality feedback. This implies that current CAs in research lack the capability to comprehensively analyze the diverse data including group task performance, group learning behaviors and group conversations, which researchers emphasize as crucial data sources to consider in diagnostic processes (Van de Pol et al., 2010; Van de Pol et al., 2014).

In terms of scaffolding strategies, the majority of CAs in existing research primarily support cognitive activities in collaborative learning (e.g. Adamson et al., 2014; Darmawansah et al., 2024; Dyke et al., 2013; Guo et al., 2024; Nguyen, 2022, 2023; Tegos et al., 2014, 2016). Common forms include retrieving and organizing relevant information (e.g. Darmawansah et al., 2024), providing learners with conceptual explanations (e.g. Nguyen, 2022), posing questions to solicit their explanations (e.g. Tegos et al., 2016), or prompting learners to engage in reasoning based on others' viewpoints (e.g. Dyke et al., 2013). Existing research has demonstrated that providing such cognitive scaffolding through CAs can effectively promote students' academic performance (e.g. Adamson et al., 2014; Guo et al., 2024; Tegos et al., 2016), system thinking (e.g. Nguyen, 2023), knowledge building (e.g. Nguyen, 2022), scientific reasoning (Dyke et al., 2013) and argumentation skills (Darmawansah et al., 2024). In a few studies, CAs demonstrate the ability to support metacognitive activities in collaborative learning (e.g. Hayashi, 2019; Martha et al., 2023; Yilmaz & Karaoglan Yilmaz, 2020). According to Yilmaz and Karaoglan Yilmaz (2020), the CA posed weekly questions to learners, aiding the CA in reflecting on the weekly collaborative activities. The study found that this CA could enhance learners' collaborative attitudes but had no impact on self-regulation skills. But referring to Martha et al. (2023), the CA timely reminded learners to engage in planning, monitoring, evaluation, and reflection during collaborative learning activities, demonstrating a positive effect on students' self-regulation skills. Very few studies have designed CAs capable of providing social scaffolding. The CA designed by Nguyen (2023) is able to encourage and affirm students, helping to foster a harmonious group collaboration atmosphere. In summary, despite past emphasis by researchers on the indispensability of cognitive, metacognitive, and social scaffolding in collaborative learning (Ouyang et al., 2022; Rojas-Drummond et al., 2013; Van de Pol et al., 2010), fewer CAs have been able to provide all three types of scaffolding. In addition,

combining the current limitations in CA diagnostic strategies, there is still significant room for improvement in the adaptive support that CAs offer.

Technological foundation of conversational agents

CA systems typically require two fundamental modules: natural language understanding (NLU) and natural language generation (NLG) (Ait Baha et al., 2023). The NLU module generally performs semantic analysis of the text (Xie et al., 2021). In the current CA systems for collaborative learning, the NLU module primarily involves calculating the similarity between learner's utterances and predefined utterances. This similarity is then utilized to identify the learner's intent (e.g. Martha et al., 2023) or assess the correctness of the learner's understanding of knowledge points (e.g. Nguyen, 2023; Tegos et al., 2016). Furthermore, the advancement of multimodal learning analytics has enabled CAs to better understand contextual situations by integrating learners' dialogue with other types of data. Some studies have explored the use of multimodal learning analytics to uncover the complex patterns within collaborative learning processes (eg. Kang et al., 2024; Ouyang et al., 2023; Xing et al., 2022). For instance, Ouyang et al. (2023) collected clickstream data and conversational data during collaborative problem-solving and used sequence analysis methods to identify typical patterns in the collaborative process. Similarly, Xing et al. (2022) have gathered behavioral log data and chat messages from students' collaborative learning activities, applying machine learning techniques to automatically analyze group engagement and uncover its connection to collaboration outcomes. These studies have laid the technological foundation for the design and development of CAs capable of understanding the collaborative learning status of groups through multimodal data.

After comprehending the collaborative learning status of groups, the CA system responds to the students through the NLG module (Xie et al., 2021). In most educational research on CAs, the NLG module of the CA system adopts the retrieval-based technique (Pérez et al., 2020), which means that the CAs retrieve predefined responses from static knowledge bases (Luo et al., 2022). Retrieval-based techniques demonstrate notable efficiency in providing swift and articulate replies (Xie et al., 2021). However, their drawbacks emerge when faced with unexpected situations, making it challenging to meet the learner's diverse and personalized needs. Recently, generative AI represented by ChatGPT achieved a groundbreaking leap in their language understanding and generation capabilities (Chen et al., 2024), which showed the tremendous potential of generative AI-based CA in providing personalized support for learners (Boudouaia et al., 2024).

Whereas many studies have applied GAI-based CAs to support individual learning (Boudouaia et al., 2024; Escalante et al., 2023; Guo & Li, 2024; Nguyen et al., 2024; Wu et al., 2024), a smaller number of researchers have begun using them to support collaborative learning. For example, Guo et al. (2024) designed a CA to enhance the quality of peer feedback and writing performance during peer assessment activities. This CA leveraged ChatGPT's strengths in language understanding and generation by analyzing the feedback students provided to their peers and offering targeted suggestions for improvement. Similarly, Darmawansah et al. (2024) employed ChatGPT to support collaborative argumentation activities, helping students formulate arguments and generate evidence. Their study found that the integration of ChatGPT significantly improved students' critical thinking awareness and the quality of their arguments. However, both of these studies that utilized GAI-based CAs to support collaborative learning have yet to explore their impact on self-efficacy, nor have they investigated students' perception on such CAs.

While researchers acknowledge the positive effects of generative AI in supporting learning, they have also pointed out its potential negative effects. A frequently emphasized concern is that learners may become overly reliant on generative AI, which can hinder their ability to think independently (Chan & Hu, 2023; Liu et al., 2024). This has been confirmed by several empirical studies. For instance, research by Niloy et al. (2024) found that ChatGPT can impede students' development of creative writing skills. Similarly, in a study by Yilmaz and Yilmaz (2023), students reported that they relied too heavily on generative AI tools to complete tasks. Another widely criticized issue is the "AI hallucination", where generative AI produces inaccurate content that could mislead learners (Barrot, 2023; Guo & Li, 2024). To address these challenges, retrieval-augmented generation (RAG) techniques present a promising solution. RAG, which integrates retrieval-based and generative-based models, allows CAs to be more precisely tailored to users' needs by drawing on external knowledge bases before generating responses (Cai et al., 2022). By retrieving relevant knowledge from a database and feeding it to the language model, RAG can enable more accurate and reliable responses (Lewis et al., 2020). In the educational context, educators could develop an education knowledge base containing scaffolding strategies and domain-specific knowledge, which could then be used by the model to generate responses aligned with particular scaffolding strategies and domain-specific knowledge. Recently, some researchers have begun advocating for the use of RAG techniques to design CAs that better support learning (Guo & Li, 2024; Lo et al., 2024). In the study by Guo and Li (2024), learners created a RAG-based CA to assist with their individual writing tasks. However, limited research to date has employed this technology to design CAs for collaborative learning.

To address current research gaps, this study, grounded in the contingent teaching model, integrated the retrieval-based model, generative AI model and RAG method to develop a CA named CollaBot. This CA can employ diagnostic strategies by combining data from group conversations, group writing behaviors, and group writing essay. Based on the diagnostic results, it could adaptively provide metacognitive, cognitive, and social scaffolding to small groups. Furthermore, the research conducted an empirical study to explore the impact of the CA on learners' learning performance and self-efficacy for writing, while also investigating learners' perceptions of the CA. This study primarily focuses on the following three research questions:

RQ1 What are the effects of the RAG-based conversational agent on students' learning performance in online collaborative writing activities?

RQ2 What are the effects of the RAG-based conversational agent on students' self-efficacy for writing in online collaborative writing activities?

RQ3 What are the students' perceptions of the RAG-based conversational agent?

Method

Participants

To investigate the effect of CollaBot, a generative AI-supported CA, the study designed an English writing practice programme to recruit participants with approval of the University

Ethics Committee. With a promise of a participation fee and a novel experience to enrich students' understanding of English writing, the study recruited 78 students (8 male, 70 female) from diverse majors at a key University in China. Using a randomized controlled experimental design, all participants were randomly assigned into the experimental condition (13 triads, with the CA) and control condition (13 triads, with the task scripts). Prior to the experimental procedures, a pretest was implemented to homogenize the two conditions and control for variables that might skew the research findings. This pretest comprised a scale measuring participants' attitudes toward collaboration applied by Rojas et al. (2022), which is essential for fostering a successful online collaborative learning environment (Ku et al., 2013). Additionally, the pretest included questions to gather information about participants' prior usage experience with collaborative writing tools and CAs. Mann–Whitney U test showed there was no distinct difference between the experimental groups and the control groups in regard to their attitude toward collaboration ($U = 675.50$, $p = .39$), previous experience with collaborative writing system ($U = 649.50$, $p = .22$), as well as previous usage about CAs ($U = 649.50$, $p = .26$). Besides, all participants also took part in a diagnostic test of English writing proficiency before the experiment, and the Mann–Whitney U test also manifested no distinct difference between the experimental and control groups ($U = 716.00$, $p = .65$). Participants had all signed a paper consent form introducing them programme details and data security promise at the beginning of the experiment, and they were told they could leave the programme at any time.

Procedures

The whole activity continued 130 min comprised with the pre-experiment, experiment, and post-experiment stages. As aforementioned, students were involved in a diagnostic test of English writing proficiency before the experiment, and they also finished a pretest with respect to their attitude toward collaboration, previous usage experience about chatbots, as well as self-efficacy for writing. To ensure the efficiency of students' collaboration, the researchers further presented them with two videos introducing the argumentative essay writing and the collaborative writing platform respectively. During the experiment period, both the experimental and control groups were required to complete a collaborative writing task within 70 min, which involved collaborative work like discussion, co-writing, and checking as well as proofreading. Nevertheless, the experimental groups in this stage did their job with the help of CollaBot, while the control group used the traditional task scripts. After the experiment, all participants would receive a post-test re-evaluating their self-efficacy for writing. Differentiated from the control groups in the post-experimental stage, members from the experimental groups would receive an additional focus group interview reflecting their perceptions towards the chatbot in this study. The whole procedure is illustrated in Fig. 1. Two experts would score the collaborative writing products with specific grading criteria after the whole process.

Learning task and environment

During the experimental stage, students' main task was to collaborate within 70 min to complete an argumentative essay about 250–350 words on a self-created collaborative writing platform. To guarantee the effectiveness of collaboration, the researchers established a new collaborative writing platform based on an open-source real-time collaborative writing tool Etherpad (<https://etherpad.org/>). As shown in Fig. 2, the platform

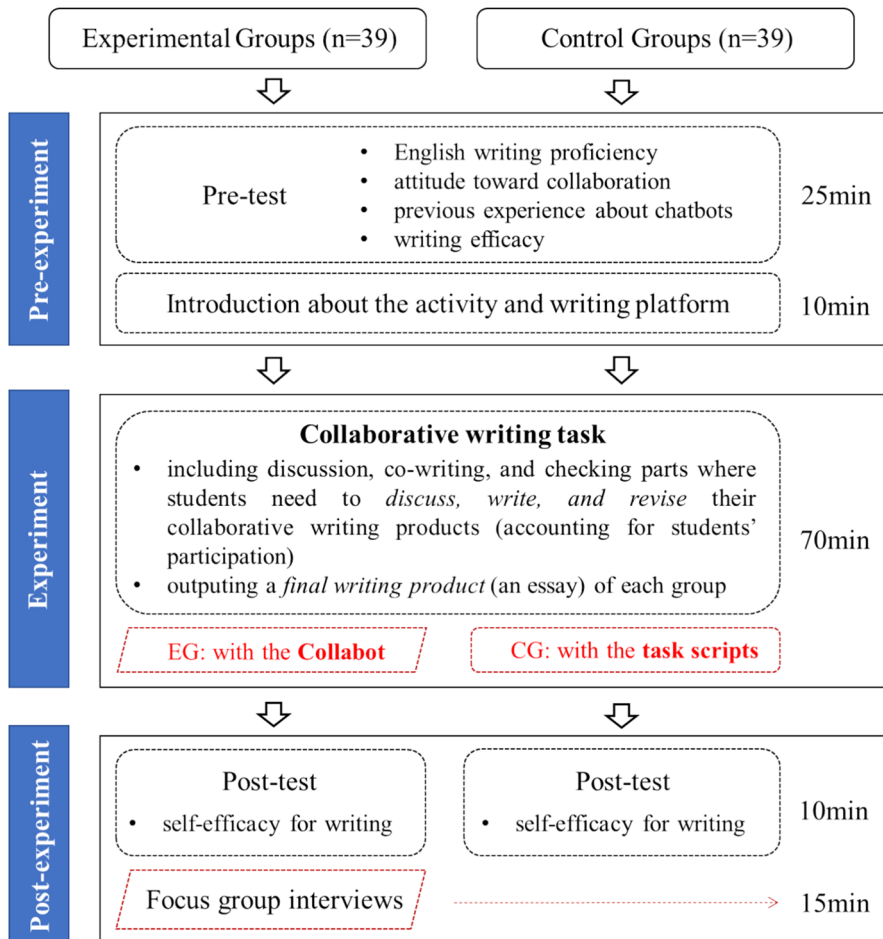


Fig. 1 Experimental procedures of this study

is mainly consisted of a collaborative editing area and a task description area. In the collaborative editing area, students could co-write their writing outline and work division in the discussion area and collaboratively complete the essay in the writing area. Besides, students could also chat in real time to share and demonstrate their opinions, identifying each other's jobs, and co-constructing their writing product in the chatting area. Each group member was represented in their unique color in both writing and chatting area, so that they could be clearly distinguished. The abundant editing tools also helped to satisfy students' basic writing needs. Apart from that, task requirements were given in the task description area, while the task scripts were accessible to control groups in terms of three aspects: organization of writing phases, instruction of essay writing skills, and common errors occurred in English writing. As for experimental groups, the researchers embedded a generative AI-based CA named CollaBot in the chatting area (Fig. 3), which worked as a pedagogical agent providing adaptive cognitive, metacognitive, social scaffoldings to small groups. In other words, regardless of

Collaborative editing area

Task description area

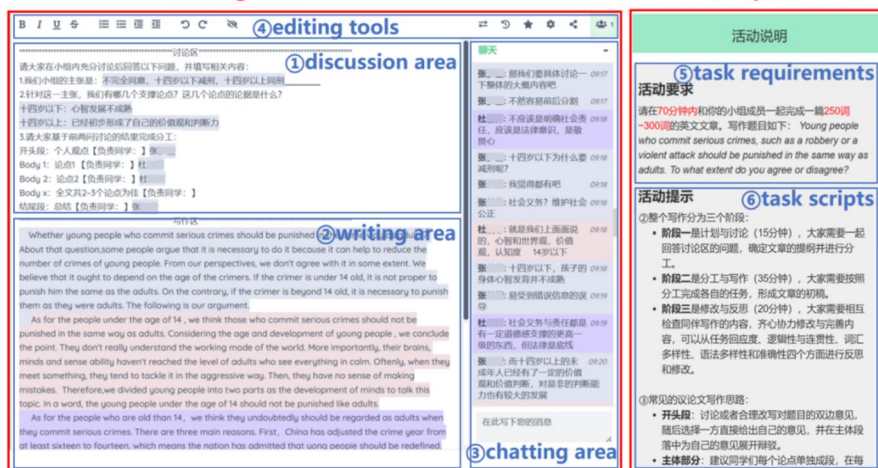


Fig. 2 Screenshot of students' co-writing environment with the task scripts (control condition)

Collaborative editing area

Task description area

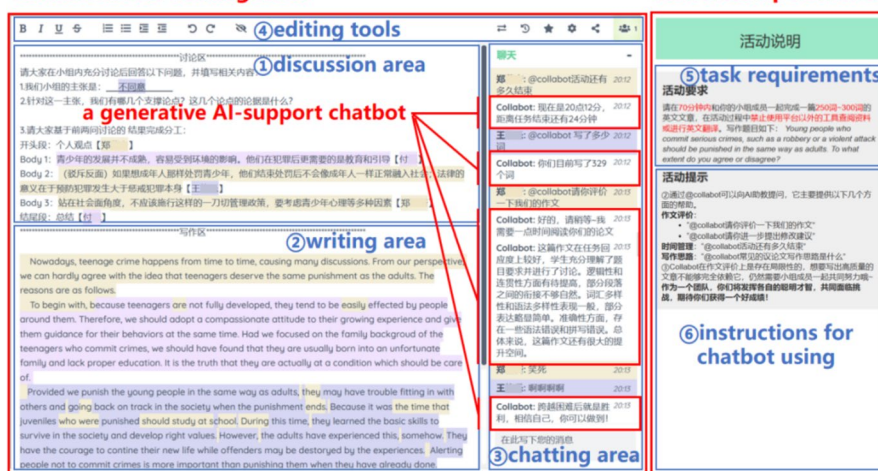


Fig. 3 Screenshot of students' co-writing environment with the CollaBot (experimental condition)

whether students were in the control or experimental groups, they all received substantial learning support through the platform during the learning process.

Agent designs

The scaffolding design of the agent

The research delineates the specific subtypes of metacognitive, cognitive, and social scaffolding provided by CA in collaborative learning scenarios, along with the timing of each

scaffolding's provision. *Cognitive scaffolding* is utilized to facilitate group knowledge construction and problem-solving (Rojas-Drummond et al., 2013; Zheng et al., 2022) and have high- or low-level of control. Low-control cognitive scaffolding involve teachers providing learners with some open-ended questions to stimulate group thinking without offering new information (Van de Pol et al., 2014). High-control cognitive scaffolding entail teachers directly providing answers to group members (Ouyang et al., 2022). This study categorizes cognitive scaffolding into two types: hint and prompt (Nguyen, 2022). Hint involves a lower level of control, offering general guidance toward problem-solving when learners seek assistance. Prompt, with a higher level of control than hint, provide further specific assistance to learners when groups cannot find solutions even with hint.

Metacognitive scaffolding primarily facilitates learners in managing group, peer, and self-learning processes. Existing research indicates a strong correlation between metacognitive activities such as planning and monitoring of the learning process within groups and group performance (Janssen et al., 2012; Su et al., 2018). Some researchers have proposed that metacognitive scaffolding in collaborative learning mainly consists of three types: planning, monitoring, and reflecting (Zheng et al., 2019). As researchers commonly posit that the process of learners regulating their learning encompasses planning, monitoring, regulating, and evaluating (Järvelä, 2011; Pintrich, 2000; Su et al., 2018), this study extends Zheng et al. (2019)'s metacognitive scaffolding framework by introducing regulating as a subcategory. This addition primarily aids individuals and groups in promptly adjusting their behaviors and plans to facilitate the seamless completion of learning tasks. The agent should diagnose the actual progress of group learners and their engagement in collaboration based on the stage of activities, in order to determine the specific type and content of metacognitive scaffolding to provide.

Social scaffolding is employed to facilitate social dynamics and foster social-emotional interactions among group members in collaborative learning processes, thereby maintaining a positive and harmonious social-emotional atmosphere within the group (Nguyen, 2023; Ouyang et al., 2022; Rojas-Drummond et al., 2013). The social scaffolding in this study is categorized into three types: acknowledge, encourage, and break the silence. The agent primarily determines the provision of social scaffolding to the group based on the group's learning progress, emotional states, and communication patterns, aiming to resolve interpersonal conflicts within the group, cultivate a conducive collaborative environment, and encourage the sharing of ideas among students. The detailed descriptions and examples of all scaffoldings can be found in the Appendix (Table 4).

The system design of the agent

According to the model of contingent teaching constructed by Van de Pol et al. (2012), diagnosing students' current learning situation and providing tailored scaffolding are key components in the scaffolding process. Inspired by this, the CollaBot system is designed with two modules: a diagnosing module and a scaffolding module, as illustrated in Fig. 4.

When groups collaborate on tasks using the online writing platform, CollaBot's diagnosing module collects real-time data on their writing behaviors, writing content, and dialogue, and stores this data in a relational database (MySQL). Subsequently, the CA conducts behavioral and textual analyses on the collected data. Behavioral analysis involves the real-time calculation of key quantitative indicators, such as the frequency of each student's chat message, the word count for each student's writing, and whether the student is actively engaged in writing. Textual analysis primarily involves assessing

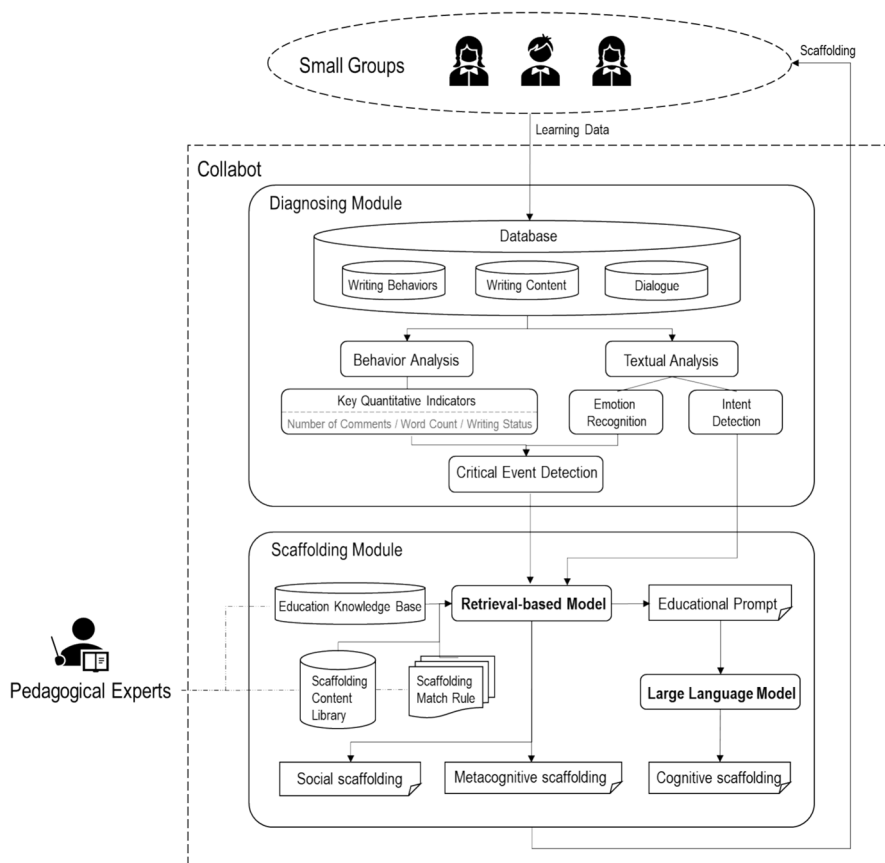


Fig. 4 Overview of CollaBot's system architecture

students' emotional states based on their statements, as well as detecting their intentions by analyzing the questions they pose to CollaBot. The results of both behavioral analysis and emotion recognition are then input into the critical event detection module, which determines whether any key events triggering scaffolding have occurred. These critical events, defined in Appendix (Table 4), represent the conditions under which scaffolding is provided. The diagnostic module sends the detected critical events and intention detection results to the scaffolding module. After that, the scaffolding module determines which type of scaffolding to trigger. For social and metacognitive scaffolding, a retrieval-based model is primarily employed. The content of scaffolding and matching rules for scaffolding are predefined by pedagogical experts. The scaffolding module matches the appropriate social scaffolding or metacognitive scaffolding based on the analysis results.

Figure 5 presents a case to clearly illustrate the workflow through which CollaBot provides metacognitive scaffolding. CollaBot continuously analyzes the data generated during group collaborative learning. When critical events are detected, such as a student failing to engage in either writing or discussion within a specified time frame, it retrieves the corresponding scaffolding content (e.g., *Regulate Participation*) from the scaffolding content

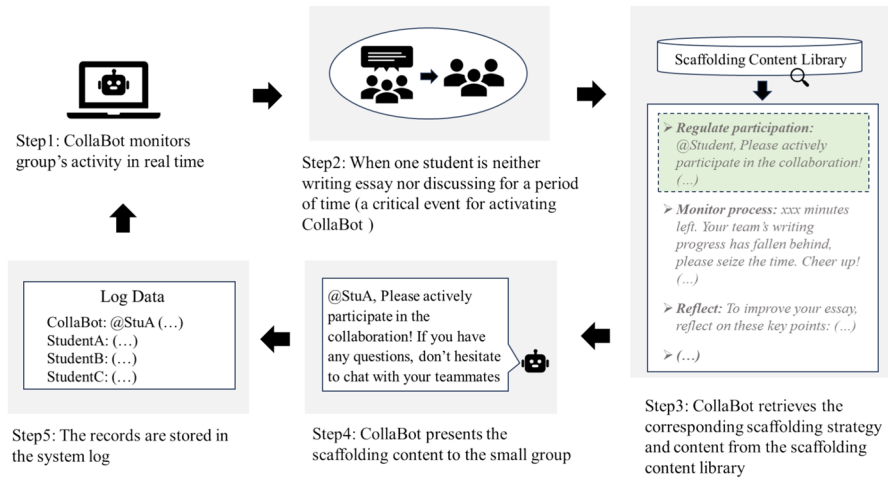


Fig. 5 Example workflow of metacognitive scaffolding

library based on the scaffolding match rule. This scaffolding content is then provided in the group chat area.

The workflow for providing social scaffolding in CollaBot is similar to that of metacognitive scaffolding. As illustrated in Fig. 6, if the diagnostic module detects negative emotions among group members through real-time analysis of their conversations, for example, a student expresses “The task is too difficult to complete.” The scaffolding module will match the corresponding *Encourage* scaffolding from the scaffolding content library. Then, the CollaBot will present the scaffolding content to students such as “Believe in yourself and believe in your team!” in response to their expressed frustration.

As for cognitive scaffolding, it is provided when students actively seek help from CollaBot. After a student poses a question, CollaBot’s diagnosing module utilizes the Sentence-BERT model to detect the student’s intent. The scaffolding module then

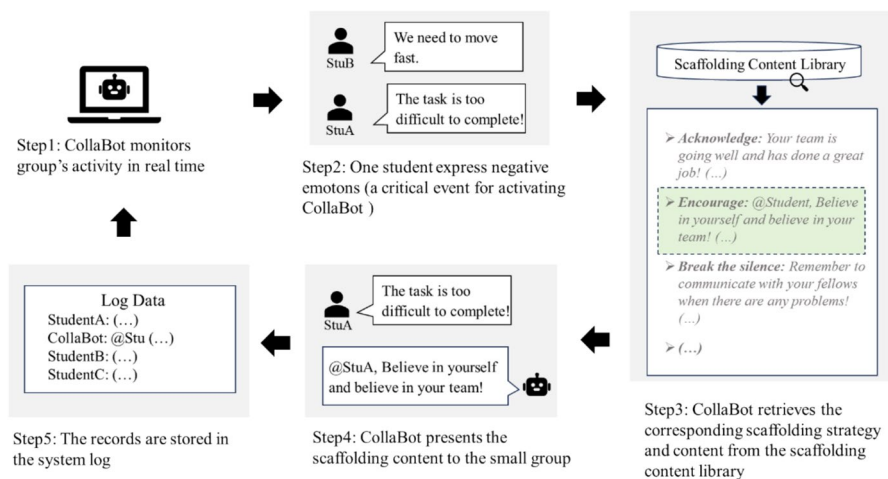


Fig. 6 Example workflow of social scaffolding

applies the RAG method, dynamically generating educational prompts and inputting them into a large language model to produce customized cognitive scaffolding. This study used the Spark (<https://xinghuo.xfyun.cn/>) as the large language model. Specifically, the scaffolding module retrieves the corresponding domain knowledge and scaffolding strategies from the educational knowledge base based on predefined rules, according to the student's intent. The educational knowledge base is designed in advance by pedagogical experts. Additionally, the scaffolding module gathers relevant student learning data (e.g., *the group's essay*) and the student's inquiry from the system database. Ultimately, domain knowledge, scaffolding strategies, students' learning data, and the posed question are automatically integrated to generate educational prompts, which are then fed into the large language model to produce the corresponding cognitive scaffolding content.

As illustrated in Fig. 7, during collaborative writing, students can request detailed revisions from CollaBot by posing a question, such as “@CollaBot, Can you give some further suggestions about our essay?” In response, CollaBot utilizes the Sentence-BERT model to accurately identify the student's intention, recognizing the need for detailed suggestions. With this understanding, it retrieves relevant domain knowledge (e.g., *the three dimensions of essay assessment rubric*) and scaffolding strategies (e.g., *Prompt, the high-control scaffolding*) from the educational knowledge base. Then, by concatenating these information with students' learning data (e.g., *group essay*), educational prompts are effectively formed. These prompts are designed to guide the output content of the large language model, ensuring that it is tailored to the unique needs of the learning group. Once the educational prompts are generated, they are then fed into the large language model. This process enables CollaBot to provide personalized and detailed revision suggestions while also encouraging reflection among the students, rather than completing the learning task for them. For example, the output might include feedback such as, “There are some inaccuracies in your expressions. For instance you can refer to this example to reflect on whether similar issues appear in your essay.” This feedback is then presented to the students as cognitive scaffoldings, with the system interface displaying the generated scaffolding content to allow students to examine and learn from it.

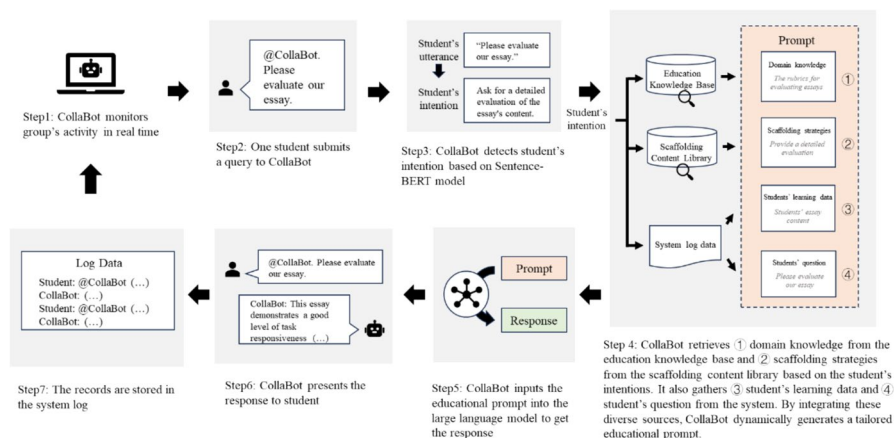


Fig. 7 Example workflow of cognitive scaffolding

Instruments

The scale of self-efficacy for writing from Bruning et al. (2013) was used both before and after the collaborative writing task, investigating three-dimensional information that are writing ideation, conventions, and self-regulation (Appendix, Table 5). According to Bruning, ideation refers to writers' beliefs about their abilities to generate ideas, for instance, "*Do I have ideas to write about, are they good ideas, and can I find the right words to express them?*" Conventions is another dimension related to a set of generally accepted standards, and sometimes is also called "mechanics". Writers may ask and think by themselves: "*Can I write and express in an appropriate way?*" As the last dimension of self-efficacy for writing, writing self-regulation is reflected in writers' confidence they can guide themselves successfully through writing's many dimensions and subtasks, with questions such as: "*Can I get started writing? Can I keep my writing going, especially when it gets difficult?*" All three dimensions provide a comprehensive consideration for investigating students' self-efficacy for writing.

Additionally, semi-structured focus group interviews were also conducted with all members in the post-experiment phase, investigating the perceptions of the experimental groups about CollaBot. The interviewer asked students their views related to the CA's scaffolding function and collected more suggestions for improvements, and the main interview questions were like: "*How have you found CollaBot helpful in the collaborative writing process?*" and "*What do you think are the shortcomings of CollaBot and what improvement do you suggest?*" With the main questions, researchers could easily refine the three main themes of the interview data that were technical usefulness, technical negativity, and students' suggestions.

Two experts then scored all groups' final products independently with grading criteria adapted from Zou and Xie (2019) after the whole experiment. A high degree of consistency between the raters was demonstrated, as indicated by a Pearson correlation coefficient of .91 between their scores. The rubric mainly contained three dimensions that are writing content, organization, and language, while sub-dimensions were also involved (Appendix, Fig. 7). Writing scores were from 1 to 5 with each corresponding to *Inadequate*, *Barely adequate*, *Satisfactory*, *Good*, and *Outstanding*. The study finally took the average of two experts as the final essay score for each group.

Data analysis

The study collected students' pre- and post-test data, collaborative writing scores, system log files, and interview data for data analysis. To answer our first research question, we took the method of Chen et al. (2023) to quantify each student's learning performance. A student's overall performance was calculated with his/her group writing score and individual participation, which occupied 70% and 30% respectively.

$$IndividualPerformance = group_{score} \times \frac{70}{100} + participation \times \frac{30}{100}$$

The individual participation was calculated according to the number of words contributed to the content, the number of words revised in the content, and the times of discussion of individual student, which can be elaborated as:

$$Participation = \left(\frac{write}{G_write} + \frac{edit}{G_edit} + \frac{chat}{G_chat} \right) \times \frac{100}{3}$$

In this equation, *write* is the number of words contributed by individual; *edit* is the number of words revised by individual; *chat* is the number of individual discussion times. Accordingly, *G_write* is the total number of words contributed by team members; *G_edit* is the total number of frequency revised by team members; *G_chat* is the total number of discussion times among team members.

To answer research question one and research question two, as the results of Shapiro–Wilk normality test showed that the sample of this study had a non-normal distribution, Mann–Whitney U test was used to compare differences between the experimental groups and the control groups in terms of students' learning performance and self-efficacy for writing. And Wilcoxon's test was used to examine the difference between the pre-tests and post-tests regarding self-efficacy for writing in both experimental groups and control group, thereby revealing whether students' self-efficacy for writing significantly improved under the support of CA or task scripts.

To further illustrate students' perceptions of the chatbot, the study conducted thematic analysis drawing on the qualitative data. Researchers identified the themes and sub-themes with repeatedly reading and confirming the interview data, and the number of students under each theme was also counted. Three themes and fourteen sub-themes were derived eventually to help better understand students' authentic opinions and suggestions.

Result

Effects of the conversational agent on students' learning performance

The results in Table 1 present the descriptive statistics and Mann–Whitney U test results for students' learning performance in the experimental and control groups. The descriptive statistics indicate that the experimental group (Mean = 69.77, SD = 5.90) generally performed better and more consistently than the control group (Mean = 65.19, SD = 9.00). Furthermore, the Mann–Whitney U test results reveal the effects of CollaBot on students' learning performance, indicating a significant difference between the experimental and control groups ($U = 557.00$, $p = .04$, $r = .58$). These findings clearly suggest that students in the experimental group, who used CollaBot, outperformed those in the control group, who relied on traditional task scripts.

Table 1 Effects of the conversational agent on students' learning performance

Groups	N	Mean	SD	U	p	Effect size $ r $
EGs	39	69.77	5.90	557.00	.04*	.58
CGs	39	65.19	9.00			

* $p < .05$ ** $p < .01$

Effects of the conversational agent on students' self-efficacy for writing

Table 2 presents the descriptive statistics for students' self-efficacy for writing in the pre-test and post-test stages for both the experimental and control groups. The results of Wilcoxon test showed that students' self-efficacy for writing was significantly improved in both experimental and control groups ($z=4.62$, $p=.00$, $r=.73$; $z=4.52$, $p=.00$, $r=.76$). In addition, the Mann–Whitney U test revealed no significant difference between the experimental and control groups in terms of their self-efficacy for writing at both the pre-test stage ($U=731$, $p=.77$) and post-test stage ($U=723$, $p=.71$).

As aforementioned, self-efficacy for writing in this study comprised three sub-dimensions that are *Ideation*, *Conventions*, and *Self-regulation*. The results of the Wilcoxon test showed that all three subdimensions significantly improved in both the experimental and control groups. In experimental groups, the effect size was largest for *Self-regulation* ($z=4.87$, $p=.00$, $r=1.02$), followed by *Conventions* ($z=2.99$, $p=.00$, $r=.42$), and then *Ideation* ($z=3.41$, $p=.04$, $r=.35$). Meanwhile, in control groups, the effect size was largest for *Self-regulation* ($z=4.02$, $p=.00$, $r=.88$), followed by *Ideation* ($z=2.01$, $p=.00$, $r=.51$), and then *Conventions* ($z=2.60$, $p=.01$, $r=.31$). Additionally, the Mann–Whitney U test showed that no significant differences were found between the experimental and control groups in the subdimensions of *Ideation* ($U=758$, $p=.98$, $r<.01$), *Conventions* ($U=713$, $p=.63$, $r=.08$), and *Self-regulation* ($U=698$, $p=.53$, $r=.10$) at the post-test stage.

Students' perceptions of the conversational agent

To answer research question 3, the study did focus group interviews facing experiment groups, thematic analysis was applied to discover students' perceptions of CollaBot. As shown in Table 3, researchers of this study discussed and concluded three major themes of students' perceptions that are technical usefulness, technical negativity, and suggestions, with 14 sub-themes were revealed as well. The n in the table represents the number of students who expressed this perception (sub-theme).

To explain them each, the first theme *Technical usefulness* reflects the contributions that students perceived of how CollaBot benefited their collaborative writing process. 8 sub-themes were included: *time management*, *group awareness*, *self-regulation*, *group regulation*, *idea inspiration*, *language support*, *writing improvement*, and *atmosphere*

Table 2 Descriptive statistics of students' self-efficacy for writing

Categories	Groups	N	Pre-test		Post-test	
			Mean	SD	Mean	SD
Ideation	EGs	39	16.69	2.69	17.72	3.23
	CGs	39	16.59	2.76	18.05	2.96
Conventions	EGs	39	18.72	1.95	19.72	2.70
	CGs	39	18.41	2.57	19.28	2.99
Self-regulation	EGs	39	20.21	3.08	23.33	3.06
	CGs	39	20.74	2.37	22.97	2.71
Total	EGs	39	55.62	6.56	60.77	7.57
	CGs	39	55.74	5.41	60.31	6.49

Table 3 Students' perceptions of CollaBot (n = 39)

Themes	Sub-themes	Descriptions	Examples	n
Technical usefulness	Time management	Students found it easier to do time management and finish their task in time	"We did a better job in time allocation and used our time more efficiently as well."	35
	Group awareness	Students paid more attention to their companions' writing process and discussion status	"I've noticed that my partners are really fast writers." "The reminder was quite helpful because it helped keep everyone discussing synchronously. We knew the engagement of our companions without the feeling of isolation."	13
	Self-regulation	Students felt the needs to regulate their own writing process and discussion status	"I always encounter that kind of situation where I don't know how to write and just stopped to think (in silence), which may be less efficient and makes myself feel my brain stops working. But when the chatbot reminded me to share ideas, I would hurry up and continue to think and talk, I felt good that way."	24
	Group regulation	Students felt the needs to regulate their collaboration process together	"We inevitably felt embarrassed to discuss at first in such an impromptu group and found it hard to start, but the chatbot helped us break that silence and facilitated us to cooperate and share more to accelerate the whole collaboration process."	18
	Idea inspiration	Students got more writing ideas during their writing process	"Personally speaking, I was not thinking of a refutation from the opposite side, so I thought it would be enough to come up with three arguments to support our own side. But with the bot's tips, we started to think about writing some counter cases in the third paragraph. It just felt like our thoughts were enriched all of a sudden!"	14
	Language support	Students learned more words, expressions, and grammars	"A lot of the expressions and vocabulary provided by the bot in the feedback were very advanced and I basically accepted to use them."	12

Table 3 (continued)

Themes	Sub-themes	Descriptions	Examples	<i>n</i>
	Writing improvement	Students learned to reflect on their writing habits and actively improve their writing skills	"I have had some bad writing habits ever since my senior high, such as using inexplicable sentences, clauses, and senseless conjunctions. With the chatbot's suggestions, I would rethink whether there were still such drawbacks in my own writing."	18
	Atmosphere intermediation	Students felt a more positive collaborative atmosphere	"I felt like the bot was pretty considerate and provided full emotional value when we were struggled with writing."	17
	Technical negativity	Anxiety	"Time was tight, and the chatbot kept reminding or pushing us, making me even more anxious. So I remember one of my companions was going to discuss some arguments in more detail, but I was the one who directly asked to enter the next phase, because the bot popped up so many times that it would really make me feel uneasy."	10
	Logical contradiction	The CA provided some contradictory suggestions	"I mentioned GDP in my writing, and it's so weird that chatbot told me to change it to its full name, but later told me to change it back to the acronym. It's like a death spiral, and I didn't listen to it in the end. I just changed it back and left it at that. That's paradoxical."	4
	Identification error	The CA misidentified students' emotions or needs	"For me, some expressions in the discussion were personal catchphrases, not meant to be complaints, but the chatbot would misunderstand and thus popped up like a bee."	10
	Ambiguous feedback	Students found it difficult to revise and polish according to the CA's feedback	"The first feedback for our writing was so insubstantial and generalized, and I didn't know where to start revising it."	18
Suggestions	Function improvement	Current functions could be further improved	"The chatbot's word-count reminder can be turned into an automatic visualized counting feature on the collaborative writing area, like what <i>Word</i> does."	26

Table 3 (continued)

Themes	Sub-themes	Descriptions	Examples	<i>n</i>
	Function extension	More functions could be involved	“Consider adding pop-ups or beeps to the page for important messages.”	8

intermediation. It is worth mentioning that 35 students endorsed that the CA helped them to better achieve *time management* during their collaboration. And a number of students also admitted the CA functioned well to facilitate their *self-regulation* ($n=24$) as well as *group regulation* ($n=18$). Additionally, the usefulness of *writing improvement* and *atmosphere intermediation* could be not ignored with 18 and 17 students referred to respectively. Some students also reported that they did pay more attention to their companions (*group awareness*, $n=13$), generated more ideas (*idea inspiration*, $n=14$), or receive *language support* ($n=12$) with the application of the CA.

The second theme *Technical negativity* mainly refines the side-effects or drawbacks of the CA, containing 4 sub-themes: anxiety, logical contradiction, identification error, and ambiguous feedback. In detail, it was intriguing to find 18 students mentioned the feedback from the CA was sometimes ambiguous and hard for them to do further revision according to the feedback (*Ambiguous feedback*), but the “studnet-say-no” feedback is actually the designed *hint scaffolding* in the CA’s scaffolding framework, aiming to avoid students’ over-dependence on CA’s direct answering. Besides, 10 students also pointed out the CA would causing *anxiety* and disrupted their work while completing the task, and the same number of students also find it confused with the chatbot’s *identification error*. The researchers also noted that 4 students mentioned a *logical contradiction* problem of the CA that caused their unpleasant experiences.

The third theme *Suggestions* is namely about suggestions students provided for the CA’s further improvement, comprising *function improvement* and *function extension*. Two-thirds of the students have put forward suggestions for the CA’s current functions ($n=26$). For example, one student said, “*The chatbot’s word-count reminder can be turned into an automatic visualized counting feature on the collaborative writing area, like what Word does.*” Another student considered, “*I think the chatbot needs to present more vivid expressions (personification) to make it more lively, and it may serve even better.*” Moreover, 8 students also gave some constructive advice for the chatbot’s function extension. For instance, they believed it’s better to “*Consider adding pop-ups or beeps to the page for important messages*” and “*Develop the chatbot’s speech function for better time savings and efficiency.*” All the suggestions were of value for the designer’s future iteration of CollaBot.

Discussion

This study developed a CA named CollaBot, integrating retrieval-based technique and generative AI model to provide real-time adaptive metacognitive, social, and cognitive scaffolding to learning groups engaged in online collaborative writing activities. Comparative analysis revealed that, in contrast to task scripts, CollaBot significantly enhanced students’ learning performance. However, some research has not consistently observed positive effects of conversational agents on student learning performance. For instance, Ait Baha et al. (2023) reported success rates in examinations of 65% for the group using chatbot, which are lower than that of group using digital content and the group using traditional learning. Similarly, Dyke et al. (2013) found that CA providing feedback had no effect on learning and even exerted a marginally negative effect on interaction intensity. The juxtaposition of these findings with the results of this study underscores the advantages of CollaBot. Similar to our study, MentorChat, developed by Tegos and Demetriadis (2017), as well as EvaluMate, developed by Guo et al. (2024), have also effectively improved student learning performance. This may be attributed

to the more diverse and adaptive scaffolding provided by these CAs compared to the CAs utilized in the aforementioned studies. Specifically, MentorChat facilitates targeted guidance by identifying knowledge points in learners' discourse, aiding in the establishment of connections between concepts (Tegos & Demetriadis, 2017; Tegos et al., 2014, 2016), a capability absent in prior CA-based interventions. Nevertheless, MentorChat is constrained by its limited language comprehension and pre-defined responses, impeding the fulfillment of learners' personalized needs. In contrast, both CollaBot and Evaluate utilize generative AI models, enabling them to generate personalized and adaptive support (Guo et al., 2024). Compared to Evaluate, CollaBot employs RAG technology, which provides dynamically generated educational prompts to generative language models. This allows CollaBot to deliver scaffolding more effectively in alignment with scaffolding strategies.

As for the second research question, the results indicate that there were no significant differences in self-efficacy for writing between the experimental and control groups in both the pre-test and post-test stages, suggesting that the impact of the CA and task scripts on students' self-efficacy for writing is comparable. Furthermore, by comparing pre-test and post-test results, the study found that both the experimental and control groups exhibited significant improvements in self-efficacy for writing. Several studies have also provided empirical evidence supporting the positive impact of similar task scripts on promoting students' self-efficacy (eg. Fraile et al., 2023; Müller & Seufert, 2018). And this study further confirms that, in collaborative learning contexts, the effect of a CA on students' self-efficacy is comparable to that of task scripts. This finding may offer insights for future educators, suggesting that, in the absence of AI tools like CollaBot, task scripts could be used to enhance students' self-efficacy for writing. Moreover, the results show that, in terms of effect size, the experimental group made greater progress in the self-regulation dimension. A similar finding was reported by Martha et al. (2023), who introduced a CA named Memo Tutor and found it to be more effective in promoting students' self-regulation skills compared to the control group. However, their study found no significant improvement in students' self-regulation compared to pre-test assessments (Martha et al., 2023). In contrast, our study observed significant improvement in students' self-efficacy for writing, including the self-regulation dimension. This underscores the advantage of CollaBot in enhancing students' self-efficacy, which may reflect the potential of generative AI technology in bolstering self-efficacy. Such perspective also corroborated in the research by (Wu et al., 2024), which found the ChatGPT-based learning aid could help students promote self-efficacy. Our study extends this finding to the context of collaborative learning.

Conducting interviews with students in experimental groups, thematic analysis was employed to comprehend students' perceptions of CollaBot. The findings revealed that most students believed CollaBot could assist them in managing their individual and group learning processes. In contrast to our findings, Yilmaz and Karaoglan Yilmaz (2020) observed that their CA enhanced students' attitudes towards collaboration but did not foster their self-regulation skills. However, both CollaBot in this study and the Memo tutor designed by Martha et al. (2023) demonstrated advantages in supporting students' self-regulation and group regulation. This disparity might stem from the fact that CA in Yilmaz and Karaoglan Yilmaz (2020)'s study was designed to facilitate learner reflection following weekly collaborative learning activities, whereas CollaBot and the Memo tutor provide real-time scaffolding during collaborative activities. Wu et al. (2024) further corroborated the positive effects of generative AI-based CAs in promoting students' regulation skills, a finding extended to the context of collaborative learning in this study. Currently, within the domain of CAs enhancing collaborative learning, very few researchers focusing on the

impact of CAs on regulation skills and regulation process (Okita & Clarke, 2021), which need be further explored.

Additionally, the results indicate that some students perceive CollaBot as providing language support and enhancing their writing skills, which partly explains why CollaBot significantly improved the learning performance of students in the experimental group. Similarly, Chan and Hu (2023) investigated college students' perception of generative AI tools, revealing that students believe generative AI tools can provide writing support and stimulate their inspiration. Besides, some students noted CollaBot's role in maintaining a positive emotional atmosphere, indicating the effectiveness of the social scaffolding designed in this study. This finding aligns with past discoveries of teachers using social scaffolding to support group learning (Ouyang et al., 2022). Overall, CollaBot indeed demonstrates sufficient potential to provide adaptive metacognitive, cognitive, and social scaffolding to collaborative groups in real-time using learning data, thereby facilitating their collaborative learning processes and outcomes. This implies that CollaBot can potentially significantly reduce the workload for teachers in monitoring and providing feedback in collaborative learning activities, especially in large classes where offering timely and tailored feedback to groups is logistically challenging and time-consuming.

Thematic analysis results indicate that students have also highlighted some of CollaBot's limitations. Specifically, some students perceive the hints provided by CollaBot as rather ambiguous and can't help them directly. This is to some extent in line with our expectations, as the original intention of designing the hint was precisely this. Given concerns raised by researchers regarding the potential over-reliance on generative AI tools among students (Chan & Hu, 2023; KikaLishvili, 2023; Mohamed, 2023), CollaBot has been intentionally designed to initially offer hints. The aim is to foster independent thinking and intra-group discussions before providing detailed guidance. Additionally, a minority of students expressed experiencing anxiety when using CollaBot. For groups encountering difficulties in task progression, CollaBot tends to offer scaffolding more frequently, potentially inducing stress among group members. However, the impact of such pressure on the collaborative learning process remains unclear and warrants further exploration in future research. Furthermore, students noted instances where CollaBot provided contradictory suggestions. This phenomenon primarily arises from CollaBot's cognitive scaffolding reliance on outputs generated by generative AI model. Despite our use of RAG technology to control the outputs of the generative AI model using prompts, their inherent limitations mean that they cannot guarantee entirely accurate outputs (Chen et al., 2024; Guo & Li, 2024).

Conclusion, limitations and future directions

Findings of this study demonstrate that CollaBot, a generative AI-based CA developed based on the contingent teaching model, effectively enhances students' learning performance and self-efficacy for writing. The study innovatively explores the potential of utilizing RAG technology and generative AI models to develop CAs which support collaborative learning. Furthermore, interviews reveal that the social, metacognitive, and cognitive scaffolding provided by CollaBot facilitate group members in regulating their own and the group's learning processes, fostering a positive and harmonious atmosphere, and aiding in the development of writing skills.

To sum up, this study offers unique insights into enhancing collaborative learning through generative AI tools, providing valuable practical implications for the design of CAs in future collaborative learning settings. In addition, the CA system architecture and the designed scaffolding framework proposed in this study possess strong transferability, allowing future researchers to adapt them to various collaborative learning contexts such as dialogic peer assessment, collaborative argumentation, and collaborative problem-solving. Additionally, this study offers insights into harnessing hybrid intelligence between teachers and GAI to support collaborative learning. The limitations of the CA, as reflected in students' perceptions, highlight the irreplaceable role of teachers. In future CA-supported collaborative learning environments, teachers will need to instruct students on how to effectively utilize CAs to support group learning and help them critically comprehend the scaffolding provided by the CA. Besides, a promising direction for future research is enhancing teacher-CA collaboration. Previous studies have designed dashboards to help teachers monitor small groups' learning progress (Li et al., 2022). Building on this, future research could explore the development of learning dashboards that visualize the timing, content, and group responses to CA-provided scaffolding. Such dashboards would enable teachers to identify instances where students struggle to comprehend the CA's scaffolding or when the CA provides inadequate support, allowing them to intervene and offer timely feedback.

However, several limitations are also supposed to be acknowledged. To begin, the current version of CollaBot is an initial prototype with room for improvement, which could be further enhanced based on the suggestions provided by students during interviews. Additionally, the cognitive scaffolding designed in this study is relatively simplistic, suggesting the potential for diversification in future iterations. Integration of pedagogical experts and the intelligence of generative AI models could offer more comprehensive support to students. Moreover, this research recruited participants for experiments rather than conducting them in real classroom settings, warranting the need for validation of its effectiveness in authentic educational environments. Furthermore, the experimental duration in this study was relatively short, highlighting the necessity for long-term investigations to explore the sustained effects of CollaBot.

Appendix

See Fig. 8 and Tables 4, 5.

As shown in Fig. 8, rubrics for grading argumentative essays were adapted from Zou and Xie (2019), emphasizing more on students' argumentative skills in the first two dimension.

	A Outstanding	B Good	C Satisfactory	D Barely adequate	F Inadequate
Content/Task Response	- A clear and fully developed position is presented which directly answers the questions. - Word length is appropriate; or within the requirements. - Ideas are highly relevant, comprehensive and fully extended. - All arguments/options/analyses very well supported and highly logical and critical, many of which show originality/creativity/sophistication; based on a good range of viewpoints and/or source materials.	- A clear and well-developed position is presented in response to the questions. - Word length is generally appropriate; or within the requirements. - Ideas are relevant, comprehensive and well extended. - All arguments/options/analyses logical and critical, with some being originality/creativity; based on a good range of viewpoints and/or source materials.	- A clear and developed position is presented. - Word length is generally appropriate; may be just below/beyond the requirements. - Ideas are generally relevant, comprehensive and extended. - Arguments/options/analyses generally logical and critical; based on a fair range of different viewpoints and/or source materials.	- A position is discernible, but the reader has to read carefully to find it. - Word length may be generally inappropriate; noticeably below/beyond the requirements. - Some main ideas are put forward, but they are limited and are not sufficiently developed and/or there may be irrelevant detail. - Arguments/options/analyses generally unconvincing or illogical.	- No relevant position can be identified, and/or there is little direct response to the questions. - Word length may be inappropriate; far below/beyond the requirements. - No relevant position can be identified. - No opinion expressed as required.
Organisation/Coherence & Cohesion	- Information and ideas are logically sequenced, and cohesion is well managed. Any lapses in coherence or cohesion are minimal. - A range of cohesive devices including reference and substitution is used flexibly. - The message can be followed effortlessly. Paragraphing is skilfully managed.	- Information and ideas are logically organised and there is a clear progression throughout the response. Occasional lapses in coherence or cohesion may occur. - A range of cohesive devices including reference and substitution is used flexibly but with some inaccuracies or some over/under use. - The message can be followed with ease. Paragraphing is used sufficiently and appropriately.	- Information and ideas are generally arranged coherently and there is a clear overall progression. A few lapses in coherence or cohesion may occur. - Cohesive devices are used to some good effect but cohesion within and/or between sentences may be faulty or mechanical due to misuse, overuse or omission. - The relationship of ideas can be followed. Paragraphing is used generally appropriately.	- Organisation is evident but is not wholly logical and there may be a lack of overall progression. Nevertheless, there is a sense of underlying coherence to the response. - Some lapses in coherence or cohesion may occur. - There may be limited/overuse of cohesive devices with some inaccuracy. The writing may be repetitive due to inadequate and/or inaccurate use of reference and substitution. - The relationship of ideas can generally be followed but the sentences are not fluently linked to each other. Paragraphing is inconsistently managed, affecting overall coherence.	- Information and ideas are evident but not arranged coherently, and there is no clear progression within the response. Many lapses in coherence or cohesion may occur. - There is some use of basic cohesive devices, which may be inaccurate or repetitive. There is inaccurate use or a lack of substitution or referencing. - Relationships between ideas can be unclear and/or inadequately marked. There is little evidence of control of organisational features.
Language	Grammar: Highly accurate; displays a very wide range of grammatical structures supporting sophisticated and clear messages; minor errors are extremely rare and have minimal impact on communication. Vocabulary: A wide range of vocabulary is used accurately and appropriately with very natural and sophisticated control of lexical features; Minor errors in spelling and word formation are extremely rare and have minimal impact on communication. Style and tone: Highly appropriate	Grammar: Accurate; displays a wide range of grammatical structures supporting clear messages; Occasional, non-systematic errors and inappropriacies occur, but have minimal impact on communication. Vocabulary: There is skilful use of uncommon and/or idiomatic items when appropriate, despite occasional inaccuracies in word choice and collocation; Occasional errors in spelling and/or word formation may occur, but have minimal impact on communication. Style and tone: Appropriate	Grammar: Generally accurate; displays an adequate range of grammatical structures; A few errors in grammar may persist, but these do not impede communication. Vocabulary: There is some ability to use less common and/or idiomatic items, though inappropriacies occur; There are a few errors in spelling and/or word formation, and they do not detract from overall clarity. Style and tone: Generally appropriate	Grammar: displays a narrow range of grammatical structures; Errors in grammar and punctuation occur, but rarely impede communication. Vocabulary: The meaning is generally clear in spite of a rather restricted range or a lack of precision in word choice; If the writer is a risk-taker, there will be a wider range of vocabulary used but higher degrees of inaccuracy or inappropriacy; There are some errors in spelling and/or word formation, but these do not impede communication. Style and tone: Generally inappropriate	Grammar: simple grammatical structures used most of the time; Grammatical errors may be frequent and cause some difficulty for the reader. Vocabulary: There may be frequent lapses in the appropriacy of word choice, and a lack of flexibility is apparent in frequent simplifications and/or repetitions. Errors in spelling and/or word formation may be noticeable and may cause some difficulty for the reader. Style and tone: Inappropriate

Fig. 8 Rubrics for grading argumentative essays

Table 4 The scaffolding provided by CollaBot

Scaffolding	Condition	Description	Example
Cognitive scaffolding			
Hint	When student ask for help	Tell students the main problems in the current article	Stu: @CollaBot. Please give some comments on our essay CollaBot: This essay demonstrates a good level of task responsiveness, with the student having a thorough understanding of the prompt and engaging in meaningful discussion. However, there is room for improvement in terms of logical coherence; transitions between some paragraphs are not sufficiently smooth. Vocabulary and grammar display moderate diversity, with some expressions appearing somewhat simplistic. In terms of accuracy, there are instances of grammatical and spelling errors. Overall, there is ample room for improvement in this essay Stu: @CollaBot. Can you give some further suggestions about our essay? CollaBot: Variety of Vocabulary: The vocabulary you use is rather limited, for example, "the young people" is repeatedly used in Body2. You can try using alternative terms such as "juveniles" or "offenders" to enhance vocabulary diversity Accuracy of Grammar: There are some inaccuracies in your expressions. For instance, in Body2, "In a short word, alerting people not to commit crimes is more important than punishing them when they have already done." could be revised to "In short, preventing crimes is more important than punishing those who have already committed them." (.....)
Prompt	When student ask for specific suggestions	Point out where errors exist in the article and give suggestions for revision	

Table 4 (continued)

Scaffolding	Condition	Description	Example
Metacognitive scaffolding			
Make plans	At the beginning of the task	Guide the group to develop a plan	CollaBot: Let's make a plan before your co-writing! For example, what's your work division, time division, and essay organization?
Monitor process	When writing progress is slow; When there is little time left	Remind group members to speed up; remind group members of remaining task time	CollaBot: xxx minutes left. Your team's writing progress has fallen behind, please seize the time. Cheer up!
Regulate participation	When there are group members who have not participated in both group discussions or writing tasks for a long time	Remind these group members to actively participate in collaboration	CollaBot: @Stu. Haven't seen you for a long time~ Remember to exchange your ideas with your team
Reflect	After entering the reflection stage	Guide students to reflect on the writing task	CollaBot: Let's review the task and gain more together!
Social scaffolding			
Acknowledge	When there are no obvious problems with the group's article or when the group task is going well	Express acknowledgement to the group	Stu: @CollaBot. Please help us to check the article CollaBot: Congratulations! No errors seem to be detected CollaBot: Your team is going well and has done a great job!
Encourage	When some group members express negative emotions, such as frustration or sadness	Encourage students to face difficulties positively	Stu: The task is too difficult to complete CollaBot: Believe in yourself and believe in your team!
Break the silence	When all members in the group have not communicated for a long time	Motivate people to communicate	CollaBot: Remember to communicate with your fellows when there are any problems!

Table 5 The scale of self-efficacy for writing

Items	1	2	3	4	5
Ideation					
1. I can think of many ideas for my writing					
2. I can put my ideas into writing					
3. I can think of many words to describe my ideas					
4. I can think of a lot of original ideas					
5. I know exactly where to place my ideas in my writing					
Conventions					
6. I can spell my words correctly					
7. I can write complete sentences					
8. I can punctuate my sentences correctly					
9. I can write grammatically correct sentences					
10. I can begin my paragraphs in the right spots					
Self-regulation					
11. I can focus on my writing for at least one hour					
12. I can avoid distractions while I write					
13. I can start writing assignments quickly					
14. I can control my frustration when I write					
15. I can think of my writing goals before I write					
16. I can keep writing even when it's difficult					

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Declarations

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